

Estimating Step Count and Walking Speed Using Smartphone IMU Data via Phyphox

Objective:

This case study aims to utilize the iPhone's built-in inertial sensors, accessed through the Phyphox application, to collect motion data, analyze acceleration patterns, and estimate both step count and average walking speed. Data analysis will be conducted using Python or MATLAB. Additionally, students will explore directional movement using magnetometer readings.

Preparatory Reading:

Before beginning the experiment, students should review the following article to understand the fundamentals of Inertial Measurement Units (IMUs) and the types of sensors they incorporate:
<https://www.advancednavigation.com/tech-articles/inertial-measurement-unit-imu-an-introduction/>

Experimental Procedure:

1. Download and install Phyphox on your iPhone.
2. Secure the phone in your pocket or attach it firmly to your body.
3. Walk a **known, measured distance** (e.g., 10 meters) at a normal pace.
4. Use the "**Acceleration**" and "**Magnetometer**" tools in Phyphox to record data.
5. Export the recorded sensor data as CSV files for further analysis.

Data Analysis Tasks:

1. Compute the **magnitude** of acceleration
2. Compute your step count (think of using peak detection)
3. Compute your average walking speed
4. Show the direction of the movement using the magnetometer